

## AI Lead Scoring and Estimated Lead Value Prediction

### Project Summary:

The AI Lead Scoring and Estimated Lead Value Prediction project is developed using **Supervised Machine Learning – Regression**. The main objective of this project is to predict the Estimated Lead Value of a potential customer based on different lead-related information.

The dataset contains information such as **website visits, time spent on the website, page views, email interactions, previous interactions, lead quality, age, annual income, sales calls, days since the last activity, lead source, and industry**.

First, the dataset was loaded using Python and Pandas. The structure of the dataset was analyzed using functions such as **describe()** and **isnull().sum()** to understand the data and identify missing values.

The missing values in the dataset were handled during the preprocessing stage. For categorical variables, missing values were replaced with "NA", while numerical missing values were handled using appropriate numerical methods such as median imputation.

Next, the variables were divided into quantitative and qualitative variables. Since Machine Learning algorithms require numerical input, the categorical variables were converted into numerical format using **One-Hot Encoding**. This converted categorical values into binary 0 and 1 variables.

After preprocessing, the independent variables were separated from the dependent variable. The target variable of this project is Estimated Lead Value, which contains continuous numerical values. Therefore, this project is classified as a regression problem.

Feature selection was then performed to identify the most important features for predicting the target variable. **SelectKBest with f\_regression** was used to select the relevant features.

The dataset was then divided into training and testing datasets. The training dataset was used to train the Machine Learning models, while the testing dataset was used to evaluate their performance. StandardScaler was used to standardize the input features so that variables with different numerical scales could be processed effectively by the algorithms.

Several supervised regression algorithms were tested, including **Linear Regression, Linear SVM, Non-Linear SVM, and Decision Tree Regression**.

The models were evaluated using the  $R^2$  score. The results obtained were:

- ❖ Linear Regression – **0.947518**
- ❖ Linear SVM – **0.089274**

- ❖ Non-Linear SVM – **0.000968**
- ❖ Decision Tree Regression – **0.661773**

Among all the tested algorithms, Linear Regression achieved the highest **R<sup>2</sup> score of 0.947518**, which is approximately **94.75%**. Therefore, Linear Regression was selected as the final model for this project.

For the final model, a **Machine Learning Pipeline** was created by combining **StandardScaler and Linear Regression**. The pipeline was trained using the prepared dataset. Combining these steps into a single pipeline ensures that the same preprocessing procedure is automatically applied when new data is provided for prediction.

The final trained model was saved using the Pickle library as **Lead\_Scoring\_Final\_Model.pkl**. This saved model can be loaded during the deployment stage and used to predict the **Estimated Lead Value of a new lead**.

For example, when new lead information is provided to the trained model, it can produce an output such as:

**Estimated Lead Value: 107640.68**

This represents the model's predicted monetary value for that particular input lead.

### **Conclusion:**

In conclusion, this project demonstrates how **Supervised Machine Learning Regression** can be used for AI-based lead scoring and lead value prediction. The complete process included data loading, data analysis, missing-value treatment, categorical data encoding, feature selection, train-test splitting, feature scaling, model training, model evaluation, model comparison, final model selection, and deployment preparation.

**Linear Regression was selected as the final model because it achieved the highest R<sup>2</sup> score of 0.947518 (94.75%) among the tested algorithms. The trained model was successfully saved and can be used to estimate the potential value of new leads.**